O & Soergel 19: Highest weight categories 2

1) Full faithfulness on Ext's cont'd.

2) Costandard objects & applications.

1) Full faithfulness on Ext's cont'd.

Let $\mathcal{C}_A$ be a highest weight category w. poset $\mathcal{T}$ & standard objects $\Delta_A(\tau), \tau \in \mathcal{T}$. Let $\underline{\mathcal{T}}$ be a poset ideal. By Sec 1.3 of Lec 18, we get a highest weight subcategory $\underline{\mathcal{C}}_A \subset \mathcal{C}_A$ w. poset $\underline{\mathcal{T}}$ & standards $\Delta_A(\tau), \tau \in \underline{\mathcal{T}}$. The following generalizes Lemma from Sec 1.4 of Lec 18.

Proposition: $\mathrm{Ext}^i_{\underline{\mathcal{C}}_A}(M,N) \xrightarrow{\sim} \mathrm{Ext}^i_{\mathcal{C}_A}(M,N)$ $\forall$ $M,N \in \underline{\mathcal{C}}_A$.

Proof:

$\forall$ $M_1 \in \underline{\mathcal{C}}_A$ admits a nonzero homomorphism from some $\Delta_A(\tau)$ & M_1 is Noetherian (cf. Sec 1.1 of Lec 18), M_1 is a quotient of a projective $\underline{P}^0$ that is a direct sum of $\underline{P}_\tau$'s, so $\underline{P}^0 \in \underline{\mathcal{C}}^\Delta_A$. Hence we have a resolution of M

$$\underline{P}_\bullet = (\ldots \to \underline{P}^{-1} \to \underline{P}^0 \to 0)$$

w. $\underline{P}^i \in \underline{\mathcal{C}}_A^\Delta$. Then $\mathrm{Ext}^i_{\mathcal{C}_A}(M,N) = H^i(\mathrm{Hom}_{\mathcal{C}_A}(\underline{P}^\bullet, N))$.

To compute $\mathrm{Ext}^i_{\mathcal{C}_A}(M,N) = \mathrm{Hom}_{\mathcal{D}^-(\mathcal{C}_A)}(\underline{P}^\bullet, N[i])$ we use a spectral sequence whose E_1-page is $\mathrm{Ext}^i_{\mathcal{C}_A}(P^j, N)$. By Sec 1.4 of Lec 18, the latter is 0 for $i \neq 0$, so the spectral sequence degenerates at E_2-page yielding the required iso. □

2) Costandard objects & applications.

2.1) Costandard objects.

Our goal here is to define objects $\nabla_A(\tau) \in \mathcal{C}_A$ called the costandard objects. Let τ' be a maximal element in $\mathcal{T}$ & $\underline{\mathcal{T}} := \mathcal{T} \setminus \{\tau'\}$. The construction is inductive: suppose we've constructed $\underline{\nabla}_A(\tau) \in \underline{\mathcal{C}}_A$ $\forall\, \tau \in \underline{\mathcal{T}}$.

Recall the equivalence $\mathcal{C}_A \to \tilde{E}\text{-mod}_{fg}$ from Sec 1.1 of Lec 18. Consider an idempotent $\varepsilon' \in \tilde{E}$ s.t. $\varepsilon' L_{\mathbb{K}}(\tau) = \mathbb{K}^{\delta_{\tau,\tau'}}$ (Sec 1.2 in Lec 18). Consider the $\tilde{E}$-module $I'_A := \mathrm{Hom}_A(\varepsilon'\tilde{E}, A)$ so that $\mathrm{Hom}_{\tilde{E}}(M, I'_A) = (\varepsilon' M)^*$ where for $N \in A\text{-mod}$, we set $N^* = \mathrm{Hom}_A(N, A)$.

The module I'_A is relatively injective: it's free over A & $\mathrm{Ext}^i_{\mathcal{C}_A}(M, I'_A) = 0$ if M is free over A (so if $A = \mathbb{K}$, then I' is genuinely injective)

Define $\nabla_A(\tau') \in \mathcal{C}_A$ to be the image of I'_A in $\mathcal{C}_A$ under $\mathcal{C}_A \xrightarrow{\sim} \tilde{E}\text{-mod}_{fg}$ so that $\nabla_A(\tau')$ is relatively injective in $\mathcal{C}_A$.

This finishes the inductive construction of $\nabla_A(\tau)$'s.

Thm: 1) $\mathrm{Ext}^i_{\mathcal{C}_A}(\Delta_A(\tau_1), \nabla_A(\tau_2)) \simeq A^{\oplus \delta_{i,0}\delta_{\tau_1,\tau_2}}$.

2) Property 1) determines $\nabla_A(\tau)$'s uniquely up to iso.

Proof:

Step 1: By Sec 1.2 of Lec 18, under the equivalence $\mathcal{C}_A \xrightarrow{\sim} \tilde{E}\text{-mod}_{fg}$, $\Delta_A(\tau') \mapsto \tilde{E}\varepsilon'$. In particular,

$$\mathrm{Hom}_{\tilde{E}}(M, I') \xrightarrow{\sim} (\varepsilon' M)^* \xrightarrow{\sim} \mathrm{Hom}_{\tilde{E}}(\tilde{E}\varepsilon', M)^* \Leftrightarrow$$

$$\mathrm{Hom}_{\mathcal{C}_A}(?, \nabla_A(\tau')) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'), ?)^* \qquad (1)$$

Step 2: If $\tau_1, \tau_2 \in \underline{J}$ we can replace $\mathcal{C}_A$ w. $\underline{\mathcal{C}}_A$ & use Sec 1 to argue by induction (on $|J|$).

So we need to consider the case when $\tau_1 = \tau'$ or $\tau_2 = \tau'$

If $\tau_1=\tau'$ & $\tau_2\neq\tau'$, then $\nabla_A(\tau_2)\in\underline{\mathcal{C}}_A \Rightarrow \mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'),\nabla_A(\tau_2))=0$;
$\mathrm{Ext}^i_{\mathcal{C}_A}(\Delta_A(\tau'),?)=0\ \forall\, i>0$ b/c $\Delta_A(\tau')$ is projective (Sec 1.2, Lec 18)

If $\tau_2=\tau'$, then by (1):

$\mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau_1),\nabla_A(\tau'))\xrightarrow{\sim}\mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'),\Delta_A(\tau_1))^*=0$. Also, $\mathrm{Ext}^i_{\mathcal{C}_A}(\Delta_A(\tau_1),\nabla_A(\tau'))=0$ for $i>0$ b/c $\nabla_A(\tau')$ is rel. injective. This finishes the proof of 1).

Step 3: Now we prove the uniqueness. By induction on $|S|$, we reduce to proving that if $M\in\mathcal{C}_A$ satisfies

$$\mathrm{Ext}^i(\Delta_A(\tau),M)\xrightarrow{\sim}A^{\oplus\delta_{i,0}\delta_{\tau,\tau'}}\ \forall\, i,\tau. \tag{2}$$

then $M\xrightarrow{\sim}\nabla_A(\tau')$.

We start by constructing a homomorphism $M\to\nabla_A(\tau')$: we use $\mathrm{Hom}_{\mathcal{C}_A}(M,\nabla_A(\tau'))\simeq\mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'),M)^*\simeq A^*\simeq A$. Take $\iota: M\to\nabla_A(\tau')$ corresponding to an invertible element of A. The composition $\mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'),M)\times\mathrm{Hom}_{\mathcal{C}_A}(M,\nabla_A(\tau'))\to\mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'),\nabla_A(\tau'))\simeq A$ is a perfect pairing, equiv.

$$\iota\circ ?:\mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'),M)\xrightarrow{\sim}\mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'),\nabla_A(\tau')) \tag{3}$$

Let $N_1=\ker\iota$, $N_2=\mathrm{coker}\,\iota$, $\underline{S}=S\setminus\{\tau'\}$. By (3) $\mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'),N_i)=0\Rightarrow N_i\in\underline{\mathcal{C}}_A$. Note $\mathrm{Hom}_{\underline{\mathcal{C}}_A}(\Delta_A(\tau),N_1)\hookrightarrow$

$\mathrm{Hom}_{\underline{\mathcal{C}}_A}(\Delta_A(\tau), M) = 0 \ \forall \tau \in \underline{\mathcal{T}} \Rightarrow N_1 = 0$. Now using (2) for $i = 0, 1$, we see that $\mathrm{Hom}_{\underline{\mathcal{C}}_A}(\Delta_A(\tau), N_2) = 0 \ \forall \ \tau \in \underline{\mathcal{T}}$. So $N_2 = 0$ □

2.2) Criterion for being standardly filtered

Here's an important application of costandards.

Thm: For $M \in \mathcal{C}_A$ that is flat over A, TFAE

1) $M \in \mathcal{C}_A^{\Delta}$
2) $\mathrm{Ext}^i_{\mathcal{C}_A}(M, \nabla_A(\tau)) = 0 \ \forall \ i > 0, \tau \in \mathcal{T}$.
3) $\mathrm{Ext}^1_{\mathcal{C}_A}(M, \nabla_A(\tau)) = 0 \ \forall \ \tau \in \mathcal{T}$

Proof:

1) $\Rightarrow$ 2) follows from Thm in Sec 2.1

2) $\Rightarrow$ 3) is a tautology.

3) $\Rightarrow$ 1): We prove this by induction on $|\mathcal{T}|$. Namely, pick a maximal element $\tau' \in \mathcal{T}$ & set $\underline{\mathcal{T}} = \mathcal{T} \setminus \{\tau'\}$. Set $H := \mathrm{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'), M)$. Since M is flat over A & $\Delta_A(\tau')$ is projective, H is flat over A hence free. Consider the map

$\psi: H \otimes_A \Delta_A(\tau') \to M$, $\psi: \varphi \otimes v \mapsto \varphi(v)$. Set

$K := \ker\psi$, $M_0 = \operatorname{im}\psi$, $M_1 = \operatorname{coker}\psi \rightsquigarrow$ SES's:

$$0 \to K \to H \otimes_A \Delta_A(\tau') \to M_0 \to 0, \quad 0 \to M_0 \to M \to M_1 \to 0$$

We note that the exact functor $\operatorname{Hom}_{\mathcal{C}_A}(\Delta_A(\tau'), \cdot)$ sends ψ to id_H so kills K & $M_1 \Rightarrow K, M_1 \in \underline{\mathcal{C}}_A$. Also $\operatorname{Hom}_{\mathcal{C}_A}(M_0, N) \hookrightarrow \operatorname{Hom}_{\mathcal{C}_A}(H \otimes_A \Delta_A(\tau'), N) = 0 \ \forall\ N \in \underline{\mathcal{C}}_A$. Consider the long exact sequence for $0 \to M_0 \to M \to M_1 \to 0$ & $R\operatorname{Hom}_{\mathcal{C}_A}(?, \nabla_A(\tau))$ w. $\tau \in \underline{T}$ getting exact sequence:

$$\operatorname{Hom}_{\mathcal{C}_A}(M_0, \nabla_A(\tau)) \to \operatorname{Ext}^1_{\mathcal{C}_A}(M_1, \nabla_A(\tau)) \to \operatorname{Ext}^1_{\mathcal{C}_A}(M, \nabla_A(\tau))$$

The 1st & 3rd terms $= 0 \Rightarrow$ the 2nd term is $0 \Rightarrow$ $\operatorname{Ext}^1_{\underline{\mathcal{C}}_A}(M_1, \nabla_A(\tau)) = 0 \ \forall \tau \in \underline{T}$. We can assume that 3) $\Rightarrow$ 1) holds for $\underline{\mathcal{C}}_A$ by induction $\Rightarrow M_1 \in \underline{\mathcal{C}}_A^{\Delta}$. It remains to show $K = 0$.

Thx to $M_1 \in \underline{\mathcal{C}}_A^{\Delta}$ & 1) $\Rightarrow$ 2) for $\mathcal{C}_A$, we get $\operatorname{Ext}^i_{\mathcal{C}_A}(M_1, \nabla_A(\tau)) = 0 \ \forall\ i > 0, \tau \in T \Rightarrow$ [long exact sequence] $\operatorname{Ext}^i_{\mathcal{C}_A}(M_0, \nabla_A(\tau)) = 0 \ \forall\ \tau \in T, i > 0$. We get an exact sequence

$$\operatorname{Hom}_{\mathcal{C}_A}(H \otimes_A \Delta_A(\tau'), \nabla_A(\tau)) \to \operatorname{Hom}_{\mathcal{C}_A}(K, \nabla_A(\tau)) \to \operatorname{Ext}^1_{\mathcal{C}_A}(M_0, \nabla_A(\tau)) = 0$$

In particular, if $\tau \in \underline{T}$, then $\operatorname{Hom}_{\mathcal{C}_A}(K, \nabla_A(\tau)) = 0$

We claim that the latter implies $K = 0$. Choose a

minimal poset ideal $\underline{J}_1$ w. $K \in \mathcal{C}_A(\underline{J}_1)$ & let $\tau_1' \in \underline{J}_1$ be a maximal element. Then, by the construction in Sec 2.1 (& their uniqueness), $(0=) \operatorname{Hom}_{\mathcal{C}_A(\underline{J}_1)}(K, \nabla_A(\tau_1')) \simeq \operatorname{Hom}_{\mathcal{C}_A(\underline{J}_1)}(\Delta_A(\tau_1'), K)^*$. But $\operatorname{Hom}_{\mathcal{C}_A(\underline{J}_1)}(\Delta_A(\tau_1'), K) \hookrightarrow \operatorname{Hom}_{\mathcal{C}_A}(P_{\tau_1'}, H \otimes_A \Delta_A(\tau'))$ – a free A-module (exercise) $\Rightarrow$ $\operatorname{Hom}_{\mathcal{C}_A(\underline{J}_1)}(\Delta_A(\tau_1'), K) = 0 \Rightarrow K \in \mathcal{C}_A(\underline{J}_1 \setminus \{\tau_1'\})$ contradicting the minimality of $\underline{J}_1$. □

2.3) Example: dual Vermas in $\mathcal{O}_0$

We sketch an explicit construction of the costandard objects in $\mathcal{O}^\Xi$ for $\Xi \in \Lambda/(W, \cdot)$. We start with the duality equivalence $\mathcal{O}_0 \xrightarrow{\sim} \mathcal{O}_0^{opp}$.

Let $\mathcal{W} \subset \mathfrak{g}$-mod denote the full subcategory of all modules M w. weight decompositions $M = \bigoplus_{\lambda \in \Lambda} M_\lambda$ s.t. $\dim M_\lambda < \infty$. Then $M^{(*)} := \bigoplus_{\lambda \in \Lambda} M_\lambda^*$ is a $\mathfrak{g}$-submodule of M^* w. $M^{(*)}_{-\lambda} = M_\lambda^*$.

Consider the automorphism θ of $\mathfrak{g}$ w. $\theta(e_i) = f_i$, $\theta(f_i) = e_i$, $\theta(x) = -x$ $\forall$ $x \in \mathfrak{h}$ (in particular, $\theta^2 = id$). Let $M^\nu := M^{(*)}$ w.

$\mathfrak{g}$-action twisted by θ. The following results are standard:

I) $M_\lambda^\vee \xrightarrow{\sim} M_\lambda^*$ $\forall \lambda$. In particular, $\bullet^\vee$ is exact.

II) $M \xrightarrow{\sim} (M^\vee)^\vee$. In particular, it's an equivalence $\mathcal{W} \xrightarrow{\sim} \mathcal{W}^{opp}$.

III) By I), if the weights of M are bounded from above, then the same holds for $M^\vee$. It follows that $L_0(\lambda)^\vee \simeq L_0(\lambda)$ $\forall \lambda \in \Lambda$.

IV) Since the objects in $\mathcal{O}_0$ have finite length, $\bullet^\vee$ restricts to an equivalence $\mathcal{O}_0 \xrightarrow{\sim} \mathcal{O}_0^{opp}$.

V) Thx to III), $\bullet^\vee$ restricts to $\mathcal{O}^\Xi \xrightarrow{\sim} \mathcal{O}^{\Xi, opp}$.

Set $\nabla_0(\lambda) := \Delta_0(\lambda)^\vee$. These are dual Verma modules.

Proposition: The objects $\nabla_0(\lambda)$, $\lambda \in \Xi$, are the costandard objects in $\mathcal{O}^\Xi$.

Proof:

Thx to the uniqueness part of the theorem, we need

to show that

$$\dim \mathrm{Ext}^i_{\mathcal{O}_0}(\Delta(\lambda), \nabla(\mu)) = \delta_{i,0}\delta_{\lambda,\mu} \qquad (4)$$

First assume $\lambda \not\leq \mu$. Consider the poset ideal $\underline{J} = \{\tau \in \Xi \mid \tau \leq \mu\}$. Since $\lambda \notin \underline{J}$ we get $\mathrm{Ext}^i_{\mathcal{O}_0}(\Delta(\lambda), N) = 0$ $\forall\, N \in \underline{\mathcal{O}}^{\Xi}$. On the other hand, $\nabla(\mu)$ has the same composition series as $\Delta(\mu)$ thx to III) above so $\nabla(\mu) \in \underline{\mathcal{O}}^{\Xi}$ finishing the proof in this case.

Next assume $\lambda < \mu$. Here we use that $\bullet^\vee$ is a contravariant self-equivalence of $\mathcal{O}^{\Xi}$ swapping $\Delta(?)$ & $\nabla(?)$ to conclude:

$$\mathrm{Ext}^i_{\mathcal{O}_0}(\Delta(\lambda), \nabla(\mu)) \xrightarrow{\sim} \mathrm{Ext}^i_{\mathcal{O}_0}(\Delta(\mu), \nabla(\lambda))$$

which is 0 by the previous case

Finally, assume $\lambda = \mu$. $\Delta(\lambda)$ is projective in $\underline{\mathcal{O}}^{\Xi}$ (for $\underline{J} = \{\tau \in J \mid \tau \leq \lambda\}$. By Sec 1.4 of Lec 18, Ext can be computed in $\underline{\mathcal{O}}^{\Xi}$. This handles the case $i > 0$.

For $i = 0$, $\mathrm{Hom}_{\mathcal{O}_0}(\Delta(\lambda), \nabla(\lambda)) \simeq \nabla(\lambda)_\lambda \simeq \Delta(\lambda)^*_\lambda \simeq \mathbb{C}$ finishing the proof $\square$